SATYA VENKATESH KOSARAJU

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SUMMARY

Analyst at Iowa State University's **Soil Machine Dynamics Laboratory** (industry partners) with an M.S. in Agricultural & Biosystems Engineering. Specialize in **Discrete Element Method (DEM) simulation**, granular flow modeling, and experimental validation of dry fertilizer spreaders - turning data into **trend analyses, user-friendly visualizations, and decision-ready insights**. Patent holder in agricultural robotics; strong across simulation, CAD, prototyping, and process automation.

EDUCATION

M.S., Agricultural & Biosystems Engineering (Thesis-based) — May 2026
Iowa State University, Ames, IA
Thesis: *"Optimization of Dry Fertilizer Spreaders: DEM-Based Material Flow Analysis and Experimental Validation."*

B.Tech., Agricultural Engineering (Undergraduate) — 2023
Vignan's Foundation for Science, Technology and Research, Guntur, India
Gold Medal - Best Outgoing Student of the Year, Vignan University.

EXPERIENCE

Analyst - Soil Machine Dynamics Laboratory, Iowa State University — July 2026
Industry partners • Ames, IA

- Conducting **quantitative and qualitative wear analysis** of agricultural tillage components - combining lab measurements with image-based computer vision (Python, OpenCV, ImageJ) for background removal, blade alignment, wear-zone segmentation, and profile comparison; developed Fiji video workflows for GoPro footage via DaVinci Resolve, extending tillage disc analysis with paired-disc comparisons and zero-intercept quadratic curve fitting for industry-facing reports.

Graduate Research Assistant - Iowa State University × Industry partners — Aug 2023 - May 2026
Soil Machine Dynamics Laboratory, Ames, IA

- **Developed, calibrated, and validated** DEM simulation workflows for urea particle flow - tuned contact parameters against **angle-of-repose** experiments; validated independently against **hopper discharge** and **impact plate** tests using percentage relative error, confidence intervals, and repeatability analysis.
- **Analyzed trends** across simulation and experimental datasets in **Python, MATLAB, and JMP**; applied **machine vision (ImageJ, OpenCV)** for image and video-based analysis; built user-friendly visualizations and reports for the industry collaborator, and automated repeatable processing to improve reproducibility.
- Designed CAD test-rig geometries in **SolidWorks** and fabricated components at the ISU SICTR Makerspace using **welding & heat work**, CNC milling, lathe, 3D printing & scanning, and laser cutting; captured measurements with **Vernier calipers, laser distance sensors, and professional photo/video** for qualitative and quantitative characterization of urea (PSD, bulk density, friction, restitution).

PATENT

Mechatronic Robotic Device for Self-Regulating and Precision Irrigation — Patent No. 202141009919 • 2021
Designed a ROS-based agricultural robotic system integrating CNN-based weed detection, targeted herbicide application, and sensor-driven irrigation control for site-specific input management.

TECHNICAL SKILLS

Simulation: DEM modeling · particle flow simulation · AcuSolve · HyperMesh · CFD fundamentals
CAD & Design: SolidWorks Fusion 360 · Creo · AutoCAD · test-rig design · prototyping
Analysis & Automation: Python · MATLAB · JMP · Excel · data visualization · trend analysis · workflow automation · machine learning
Machine Vision & Media: Machine vision · ImageJ · OpenCV · professional photo & video capture · DaVinci Resolve
Mechanical & Fabrication: Welding & heat work · metal cutting · CNC milling · lathe · 3D printing & scanning · laser cutting
Measurement & Instrumentation: Vernier calipers · laser distance sensors · DAQ · load/force sensors · soldering · electronics

PRESENTATIONS, AWARDS & AFFILIATIONS

Presentation: *"Developing Digital Twins of Cleaning Shoe Separation of Corn Biomass Particles,"* CoMFRE Conference, Iowa State University, 2023.

Leadership (Iowa State University): Chair, **ISTVS Student Chapter**; Organizer, **Soil Compaction Day**; **Farm Progress Show Ambassador**; **Engineering Career Services Ambassador**. Also, **World Food Prize Ambassador**, Des Moines (2024).
Recognition: Gold Medal - Best Outgoing Student of the Year, Vignan University, 2023. **Memberships:** ASABE; ISTVS; CoMFRE.